

Vel Tech Group of Institutions

Name: Gontu Yaswanth

Email: vtu33434@veltech.edu.in

Roll no: Vtu33434

Phone: null

Branch: VTU

Department: CSE

Batch: 2028

Degree: B.Tech - Computer Science and Engineering

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2028_NeoColab_Competitive Coding II_L8

VTU_Week 6_COD

Attempt : 1

Total Mark : 50

Marks Obtained : 50

Section 1 : Coding

1. Problem Statement

Krish is participating in a coding competition and has been given an array of integers. His task is to sort the array using the Counting Sort algorithm.

Your goal is to help Krish by implementing the countSort method to sort the array in non-decreasing order.

Input Format

The first line of input consists of an integer N, representing the number of elements in the array.

The second line of input consists of N space-separated integers, representing the elements of the array.

Output Format

The output prints a single line containing N space-separated integers representing the sorted array.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 5

4 2 2 8 3

Output: 2 2 3 4 8

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
int main() {
    int n;
    scanf("%d", &n);

    int arr[20], count[101] = {0};

    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
        count[arr[i]]++;
    }

    for (int i = 0; i <= 100; i++) {
        while (count[i] > 0) {
            printf("%d ", i);
            count[i]--;
        }
    }

    return 0;
}
```

Status : Correct

Marks : 10/10

2. Problem Statement

Raj, a keen problem solver, has come across an intriguing numerical puzzle that involves sorting numbers in ascending order. Eager to crack the code, he decides to implement the Radix Sort algorithm.

Your task is to assist Raj in designing a program to solve this numerical puzzle.

Input Format

The first line contains an integer n , denoting the number of elements to be sorted.

The second line consists of n space-separated integers, representing the numbers to be sorted.

Output Format

The output prints the array after it has been sorted using the Radix Sort algorithm, separated by a space.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 5

543 986 217 765 329

Output: 217 329 543 765 986

Answer

```
// You are using GCC
#include <stdio.h>
```

```
int getMax(int arr[], int n) {
    int max = arr[0];
    for (int i = 1; i < n; i++)
        if (arr[i] > max)
            max = arr[i];
    return max;
}
```

```
void countSort(int arr[], int n, int exp) {  
    int output[20], count[10] = {0};
```

```
    for (int i = 0; i < n; i++)  
        count[(arr[i] / exp) % 10]++;
```

```
    for (int i = 1; i < 10; i++)  
        count[i] += count[i - 1];
```

```
    for (int i = n - 1; i >= 0; i--) {  
        int idx = (arr[i] / exp) % 10;  
        output[count[idx] - 1] = arr[i];  
        count[idx]--;
```

```
    }  
    for (int i = 0; i < n; i++)  
        arr[i] = output[i];  
}
```

```
void radixSort(int arr[], int n) {  
    int max = getMax(arr, n);
```

```
    for (int exp = 1; max / exp > 0; exp *= 10)  
        countSort(arr, n, exp);  
}
```

```
int main() {  
    int n;  
    scanf("%d", &n);
```

```
    int arr[20];
```

```
    for (int i = 0; i < n; i++)  
        scanf("%d", &arr[i]);
```

```
    radixSort(arr, n);
```

```
    for (int i = 0; i < n; i++)  
        printf("%d ", arr[i]);
```

```
    return 0;
```

```
}
```

Status : Correct

Marks : 10/10

3. Problem Statement

Amit, an e-commerce manager, is working on an inventory management system to track the top-selling products. Given the sales data of various products, he wants to quickly determine the Kth best-selling product to make informed restocking decisions.

Help Amit implement Heap Sort to efficiently find the Kth best-selling product from the given sales data.

Input Format

The first line of input consists of an integer n , representing the number of products.

The second line consists of n space-separated integers, representing the sales data of each product.

The third line contains an integer K , representing the rank of the product to retrieve.

Output Format

The output prints the Kth best-selling product based on the given sales data.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5
10 20 15 5 25
2

Output: 20

Answer

```
// You are using GCC
#include <stdio.h>
```

```
void heapify(int arr[], int n, int i) {  
    int largest = i, l = 2*i + 1, r = 2*i + 2;
```

```
    if (l < n && arr[l] > arr[largest])  
        largest = l;
```

```
    if (r < n && arr[r] > arr[largest])  
        largest = r;
```

```
    if (largest != i) {  
        int temp = arr[i];  
        arr[i] = arr[largest];  
        arr[largest] = temp;  
        heapify(arr, n, largest);  
    }  
}
```

```
void heapSort(int arr[], int n) {  
    for (int i = n/2 - 1; i >= 0; i--)  
        heapify(arr, n, i);
```

```
    for (int i = n - 1; i > 0; i--) {  
        int temp = arr[0];  
        arr[0] = arr[i];  
        arr[i] = temp;  
        heapify(arr, i, 0);  
    }  
}
```

```
int main() {  
    int n, k;  
    scanf("%d", &n);
```

```
    int arr[100000];
```

```
    for (int i = 0; i < n; i++)  
        scanf("%d", &arr[i]);
```

```
    scanf("%d", &k);
```

```
    heapSort(arr, n);
```

```
printf("%d", arr[n - k]);  
    return 0;  
}
```

Status : Correct

Marks : 10/10

4. Problem Statement

Ragavi, a computer scientist, is working on efficient text processing for search engines. She has a list of words in random order and needs to sort them in lexicographic order using Heap Sort to improve search performance.

Help Ragavi implement Heap Sort to arrange the words in lexicographic order efficiently.

Input Format

The first line consists of an integer N denoting the number of strings.

The next line consists of N space-separated strings.

Output Format

The output displays N space-separated sorted strings in lexicographic order.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 10

banana pineapple mango chikoo jack star guava lemon tomato orange

Output: banana chikoo guava jack lemon mango orange pineapple star tomato

Answer

```
// You are using GCC  
#include <stdio.h>
```

```
#include <string.h>
```

```
void swap(char a[], char b[]) {  
    char temp[50];  
    strcpy(temp, a);  
    strcpy(a, b);  
    strcpy(b, temp);  
}
```

```
void heapify(char arr[][50], int n, int i) {  
    int largest = i;  
    int l = 2*i + 1;  
    int r = 2*i + 2;  
  
    if (l < n && strcmp(arr[l], arr[largest]) > 0)  
        largest = l;  
  
    if (r < n && strcmp(arr[r], arr[largest]) > 0)  
        largest = r;  
  
    if (largest != i) {  
        swap(arr[i], arr[largest]);  
        heapify(arr, n, largest);  
    }  
}
```

```
void heapSort(char arr[][50], int n) {  
    for (int i = n/2 - 1; i >= 0; i--)  
        heapify(arr, n, i);  
  
    for (int i = n - 1; i > 0; i--) {  
        swap(arr[0], arr[i]);  
        heapify(arr, i, 0);  
    }  
}
```

```
int main() {  
    int n;  
    scanf("%d", &n);  
  
    char arr[20][50];
```

```
for (int i = 0; i < n; i++)
    scanf("%s", arr[i]);

heapSort(arr, n);

for (int i = 0; i < n; i++)
    printf("%s ", arr[i]);

return 0;
}
```

Status : Correct

Marks : 10/10

5. Problem Statement

You're organizing a gift-giving event where participants exchange gifts of varying values. To ensure fairness, you use the Bucket Sort algorithm. You gather the participants' gift values and initialize a count array.

Then, you iterate through the values, updating the count array. Next, you sort the values in descending order by iterating through the count array. Finally, you display the sorted values, ensuring fairness in the gift exchange.

Input Format

The first line contains an integer N, representing the number of gift values.

The second line contains N space-separated integers representing the gift values brought by the participants. Each gift value is an integer between 1 and 999, inclusive.

Output Format

The output displays a single line containing N space-separated integers representing the sorted gift values in descending order.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 6

156 378 453 278 281 987

Output: 987 453 378 281 278 156

Answer

// You are using GCC

#include <stdio.h>

```
int main() {
    int n;
    scanf("%d", &n);

    int count[1000] = {0}, x;

    for (int i = 0; i < n; i++) {
        scanf("%d", &x);
        count[x]++;
    }

    for (int i = 999; i >= 1; i--) {
        while (count[i] > 0) {
            printf("%d ", i);
            count[i]--;
        }
    }

    return 0;
}
```

Status : Correct

Marks : 10/10